energy&fuels

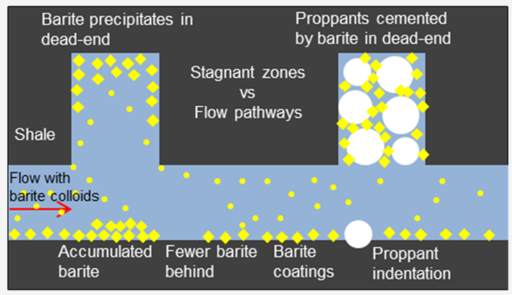
pubs.acs.org/EF Article

Influence of Flow Pathway Geometry on Barite Scale Deposition in Marcellus Shale during Hydraulic Fracturing

Wei Xiong,* Hang Deng, Johnathan Moore, Dustin Crandall, J. Alexandra Hakala, and Christina Lopano

Cite This: *Energy Fuels* 2021, 35, 11947−11957 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: Barite scaling is a common problem in the shale gas industry. Barite precipitation due to rock-fluid interactions has been studied intensively in static and flow-through experiments, but the impact of flow pathway geometry on barite scaling remains a question. The complex fracture passages can lead to local concentrated geochemical interactions, resulting in spatially variable scaling distribution. In this study, designed patterns with channels and holes were milled in two Marcellus shale cores to represent the main flow pathways, where the slow flow travels from the inlet to the outlet, and near-stagnant zones, where the fluid is trapped. Hydraulic fracturing fluid, created using synthetic produced water and Monongahela River water as base fluid, was injected at 0.02 mL/min into the cores at 66 °C, a core pressure of 12.4 MPa (1800 psi), and a confining pressure of 13.8 MPa (2000 psi) for 28 days. For the core with both channels and holes, barite coatings formed in the channels and the holes, with thicker barite accumulations occurring on the non-patterned half of the core adjacent to the holes on the milled half. For the core with only holes, proppants were added to prop up the main flow pathways. Barite formed on the propped fracture surface. The proppants in the holes were covered with and glued together by barite. Such barite-proppant conglomerates were not found in the main propped fracture. These barite-proppant conglomerates may block transport pathways as well as the barite coatings.

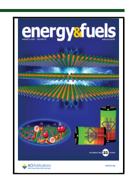
1. INTRODUCTION

Hydraulic fracturing is a technology that involves injecting high pressure water-based fluid into low-permeability formations to induce fractures, which enhances reservoir permeability and extractability of oil and gas.[1−3] Natural gas production from shale formations has boomed over the past two decades due to the development of hydraulic fracturing technology in the United States.[4] The main issues during shale gas production include low recovery rates, 20−30% compared to the estimated resource,[5,6] and the fast decrease of gas production within several months to years after the initial extraction.[6,7] The sharp decline in gas production is likely the result of many factors, including a reduced pressure gradient as the reservoir is depleted over time.[6] Additionally, mineral scaling as a result of rock-fluid geochemical interactions may block pores and fractures in shale, as well as pipelines,[8] thereby impacting flow. Among the shale plays in the United States, the Marcellus Shale is the most geographically extensive, stretching across Pennsylvania, West Virginia, and into eastern Ohio and western New York.[9] Mineral scale precipitation, especially barite scaling, is a common problem in Marcellus shale gas production.[10−14]

Monongahela River water is widely used as base fluid for hydraulic fracturing in shale reservoirs in Pennsylvania and West Virginia.[15] Monongahela River water contains high concentrations of SO_4^{2-} (50−300 mg/L)[16] compared to typical fresh natural water (20 mg/L)[17] due to fuel extraction activities.[16] Produced water reuse is common in Marcellus shale gas production sites.[18] The produced water contains high total dissolved solids (TDS) including Ba^{2+}, Sr^{2+}, Na^{+}, Cl^{-}, Mg^{2+}, Fe^{2+}, and Ca^{2+} as major components.[19−22] The hydraulic fracturing fluid (HFF) can contain 80% or more Monongahela River water that is blended with reused Marcellus produced water.[23,24] Mixing the Monongahela River water with produced water can result in an injection fluid supersaturated with respect to barite, leading to barite scaling once injected into the reservoir. Some operators remove pre-precipitated barite by filtration in above-ground facilities, but a study showed that barite would still precipitate due to rock-fluid interactions with the supersaturated HFF after injection.[24] As long as produced water is reused with Monongahela River water for hydraulic fracturing in this region, barite scaling can be a potential problem.

Depending on the location of barite scaling, transport pathways may be inhibited or even blocked for shale gas extraction. Barite precipitation in shale due to rock-fluid

Received: May 6, 2021
Revised: July 8, 2021
Published: July 21, 2021



https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

Table 1. Semi-Quantitative Mineralogical Composition (weight %) of the Marcellus Shale Core[31]

sample ID	location	quartz (%)	muscovite/illite (%)	chlorite (%)	calcite (%)	pyrite (%)	barite (%)
MIP3H-36	Monongalia CO, WV, USA	37	48	8	1	3	n/a

interactions also can affect other observed mineral reactions that impact transport pathways from the shale matrix to primary fractures. Reaction fronts containing an Fe(III) layer as a result of pyrite oxidative dissolution were observed at rock-fluid interfaces due to oxygen diffusion into the shale matrix.[25] Barite coatings observed on the Marcellus shale fracture surfaces[15,26,27] may reduce oxygen transport, thereby inhibiting further pyrite oxidation.[28] Alternatively, shale mineral dissolution by the HFF may increase porosity and enhance hydrocarbon transport.[29] The presence of barite coatings could reduce direct contact between rock and fluid, inhibiting further dissolution reactions. In a diffusion experiment with barium and sulfate injected from opposite ends, barite precipitated in larger microfractures within the rock, causing localized clogging, and then further grew into nanopores and smaller fractures, which are diffusion-controlled regions.[30] During the hydraulic fracturing process, the main flow pathways are the places where the flow is laminar or turbulent and the stagnant zones in shale may be diffusion-dominant. How different flow pathways impact barite scaling in hydraulically fractured shale remains a question.

In this study, Marcellus shale cores with engineered patterns representing the main flow transport pathways and stagnant zones were used for flow-through experiments at representative reservoir temperature and pressure. The designed patterns enabled two pairs of comparison: (1) channeled fracture without proppants vs propped fracture and (2) main flow regions vs dead-end holes. In addition to experimental analysis, reactive transport modeling was conducted to illustrate barite scaling in the flow field of the designed pathways. The objectives of this study include (1) locating barite precipitation in the main flow-pathways and stagnant zones, (2) investigating the impact of flow pathways on barite extent and distribution, and (3) identifying implications for barite control and shale gas production.

2. EXPERIMENTAL METHODS

2.1. Marcellus Shale with Engineered Flow Pathways. Two Marcellus shale sub-cores (3.81 cm in diameter and 10.16 cm in length) from a 2279.4 m depth were extracted from a larger parent core taken from the Marcellus Shale Energy and Environment Laboratory (MSEEL). The mineral composition of this parent core was analyzed using a X-ray diffractometer (XRD) and reported previously (Table 1).[31]

The two cores were cut into half cylinders. Two patterns were milled on one half of each core (Figure 1). The M-core pattern geometry (Figure 1a) included two channels (~0.5 mm depth and ~4 mm width) and three dead-end holes (~2 mm diameter and ~3 mm depth) evenly distributed on each channel. The channels served as the main flow pathways, and no proppants were needed to keep the channels open. The flow in the channels can be viewed as simple laminar flow without obstacles. The P-core pattern geometry (Figure 1b) included the dead-end holes at the same locations but without the channels. A layer of Innoprop 40/60 mesh quartz proppants was added in between the half cores to maintain a propped fracture as the main flow pathway. The propped fracture width was in the range of 50−700 μm based on post-reaction X-ray computed tomography (CT) images. The presence of proppants added tortuosity to the main flow pathway and may impact the reactive transport of the fluid. The dead-end holes were the stagnant zones where fluid has very limited mobility. The flat surfaces of the half cylinders were polished with super fine (400 grit) sand paper after milling. The patterned half cylinders were placed on top the non-patterned half to minimize the gravity influence during the flow-through experiment.

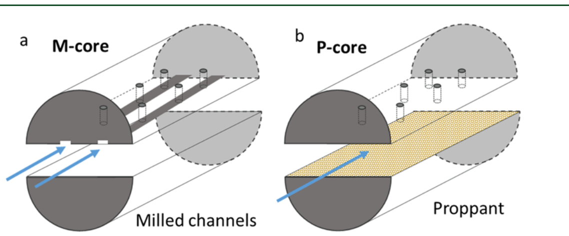


Figure 1. Patterns that simulate the main flow pathway and stagnant zones. M-core: the core with milled channels, P-core: the core that was propped with proppants.

2.2. Hydraulic Fracturing Fluid (HFF). The HFF used the same synthesized recipe as our previous studies[15] with fresh Monongahela River water (Mon water) as base water. Mon water was collected from the near surface of the river on June 10, 2019, and kept at 4 °C. Major species concentrations of the Mon water are listed in Table 2. The hydraulic fracturing chemicals (Table 3)[15,26,32] and salts (Table 4)[15,26,33] were added to the Mon water 2 h prior to the start of the experiment. This formulation simulates the concentrated mixture where produced water is blended with fresh Mon water for hydraulic fracturing in the field.

2.3. Flow-Through Experimental Setup. The same flow-through experimental system (Figure 2)[15] as our previous studies was used in the current study. The cores were placed in rubber sleeves in separate core holders with the patterned side on the top. A 13.8 MPa (2000 psi) confining pressure was added on the core holder via a syringe pump with deionized water, and 12.4 MPa (1800 psi) core pressure was set. The core holder temperature was maintained at 66 °C, which is representative of the downhole temperature. The HFF was purged with N_2 gas to limit O_2 in the fluid in a reservoir bottle. However, the system was not in a strictly anoxic environment, and the HFF contains oxidative chemicals such as persulfate. The fluid was pumped into each core at a slow rate of 0.02 mL/min. Because the experiment simulated the shut-in period, the flow rate was set as slow as possible while generating enough effluent for analysis every few days. The effluent was stored in the backpressure syringe pump and sampled every few days of the experiment. For the experimental initial setup, the HFF was pumped in quickly at 2 mL/min for 5 min to flush through the entire flow-through system. Then the flow rate was reduced to the 0.02 mL/min experimental rate while increasing the pressure and temperature to experimental conditions. Based on our previous experiments investigating the flow rate impact on barite precipitation using the same system,[15] the dissolved mass amount from the initial 5 min flush-through was negligible compared to the entire reaction time. Both cores reacted for 28 days. The system took approximately 2 h to stabilize, and the effluent from this period was removed. The reaction time was considered as the time after the start-up period.

2.4. Analytical Methods. Inductively coupled plasma mass spectrometry (ICP-MS, PerkinElmer Nexion 350D), optical emission spectrometry (ICP-OES, PerkinElmer DV 7300), and ion chromatography (IC, DIONEX ICS-5000, Thermo Scientific) were used to measure the aqueous species concentrations. Inorganic carbon and non-purgeable organic carbon were measured by TOC analysis (TOC-L, Shimadzu). Mineral saturation indices were calculated using Geochemists' Workbench (GWB) 10.0 based on measured species concentrations and pH in the solution at 66 °C, with the GWB-provided thermo_pitzer database, which is suitable for high-salinity

https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

Table 2. Mon River Major Cation and Anion Concentrations

species	Mg	Ca	K	Na	chloride	nitrate	sulfate
concentration, mg/L	6.22	24.37	1.46	11.63	5.69	2.54	59.43

Table 3. Hydraulic Fracturing Fluid Chemical Additives[15,26,32]

HFF ingredients	purpose	weight %
HCL (37%)	perforation cleaner	0.25
WGA-15 L	gelling agent	0.15
WCS-63ILC	clay stabilizer	0.106
WFR-6ILA	friction reducer	0.049
ammonium persulfate	breaker	0.02
glutaraldehyde	biocide	0.019
potassium hydroxide	pH adjuster	0.014
potassium carbonate	pH adjuster	0.012
revert flow	surfactant	0.0079
ethylene glycol	scale inhibitor	0.0045
citric acid	iron control	0.0034
boric acid	cross-linker	0.002
ethanolamine	cross-linker	0.0014
WAI-25ILC	corrosion inhibitor	0.0013

Table 4. Chemicals Added to Simulate Produced Water Composition[15,26,33]

produced water ingredients	weight %
boric acid	0.002
potassium carbonate	0.024
barium chloride dihydrate	0.046
potassium chloride	0.022
strontium chloride	0.14
ammonium chloride	0.016
sodium bromide	0.018
calcium chloride dihydrate	0.74
magnesium chloride	0.19
sodium chloride	1.67
sodium sulfate	0.00003
sodium bicarbonate	0.015

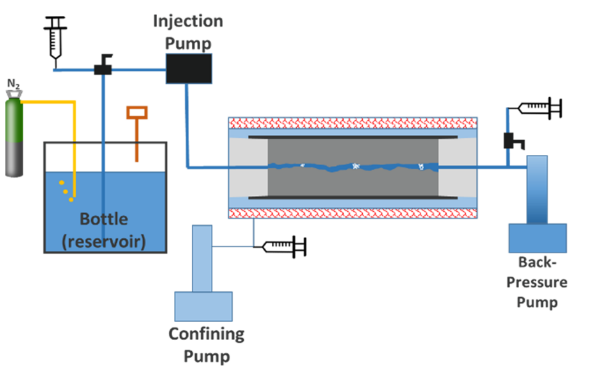


Figure 2. Schematic of the high pressure and high temperature flow-through experimental system.[15]

solution. Pre-reaction and post-reaction maps of the cores were collected via a back-scatter detector (BSE) on a scanning electron microscope (SEM, Quanta 600F, FEI). Point analyses with energy dispersive X-Ray spectroscopy (EDS, Oxford) were performed. The reacted surface and bottom of the holes were also imaged via digital microscopy (Olympus). X-ray computed tomography (CT, M5000 Industrial Computed Tomography System, North Star Imaging) was performed with the cores inside the sleeve at ambient conditions before and after reaction. The voxel resolution of the CT images was 17 μm for pre-reaction scans and 18.8 μm for the post-reaction scans. Since the post-reaction scans were taken after depressurization, the fracture aperture derived from the CT data is likely slightly larger than it was during in situ conditions. CT images were processed using ImageJ.[34] Pre-reaction X-ray fluorescence (XRF) maps of selected elements on the core fracture surfaces were taken at the Stanford Synchrotron Radiation Lightsource (SSRL) with Beamline 10-2.

2.5. Reactive Transport Modeling. Reactive transport modeling was performed using a coupled CrunchTope-Comsol framework.[35] In this modeling framework, flow is solved within Comsol using the Brinkman solver. A 2D cross section (Figure 3) along the milled channel was selected, which considers both the main flow channel and the dead-ends.

The flow field is then imported into CrunchTope for solving the transport and reactions. A quasi-steady state approach is adopted: flow calculation is updated at times based on the geometry change caused by mineral reactions. The flow field simulations used an extremely fine finite element mesh generated by the internal mesh generator of Comsol with a constant average velocity ($\overline{u}$) of 8.3333×10^{-5} m/s (which is calculated based on the experimental flow rate and the fracture cross section area) at the inlet and fixed atmospheric pressure at the outlet. The results were mapped to the Cartesian grid cells in CrunchTope, which has a resolution of 400 μm along the flow direction, and a resolution of 50 and 100 μm perpendicular to the major flow direction in the channel and in the dead ends, respectively. The reactive transport simulations used mineral composition and influent chemistry reported in this study and the previously reported reaction networks[36] developed for Marcellus shale and hydraulic fracturing fluid interactions. Specifically, barite was allowed to precipitate on the fracture surface.

3. RESULTS AND DISCUSSION

3.1. Fluid Chemistry. The influent and effluent of the M-core (with milled channels) and the P-core (with proppants), pH, and concentrations are displayed in Figure 4, and other species are included in the Supporting Information (Figures S1 and S2). Sulfate concentration in Mon water was ~60 mg/L and decreased to ~40 mg/L in the injection fluid (Figure 4a), indicating that approximately 33% of the sulfate precipitated from solution during the preparation of HFF. The relatively high sulfate concentration (50−300 mg/L) in the Mon River is due to regional mining and fossil fuel extraction activities.[16] Acid mine drainage is formed when pyrite, which is a common mineral in Marcellus shale, is exposed to air and water during extraction. The inorganic S can be converted to sulfate and H_2SO_4 with the overall reaction $2FeS_2 + 7O_2 + 2H_2O \rightarrow 2FeSO_4 + 2H_2SO_4$.[37] Acid mine drainage from coal fields causes high concentrations of free sulfuric acid and sulfate in most streams in the west Pennsylvania region.[38] When sulfate-containing Mon water and Ba-rich produced water for HFF reuse are mixed, barite forms even with a scale inhibitor.[15] Small round barite seeds in the injection fluid were observed from our previous study using the same HFF recipe.[15] The presence of these barite seeds in the HFF can facilitate barite precipitation once injected into the reservoir.

After the fluid was pumped into the cores at elevated temperature and pressure, Ba concentration decreased drastically in both the effluents (Figure 4b), indicating barite precipitation inside the cores. Ba^{2+}(aq) in the HFF is a result of the reuse of the high TDS produced water. A recent study

https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

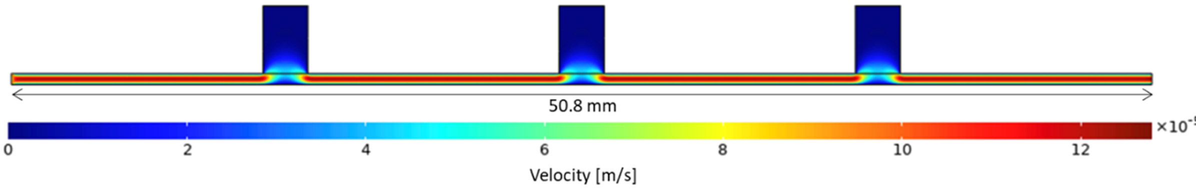


Figure 3. 2D cross section of the fracture used for the simulations, and the velocity magnitude of the original flow field from COMSOL.

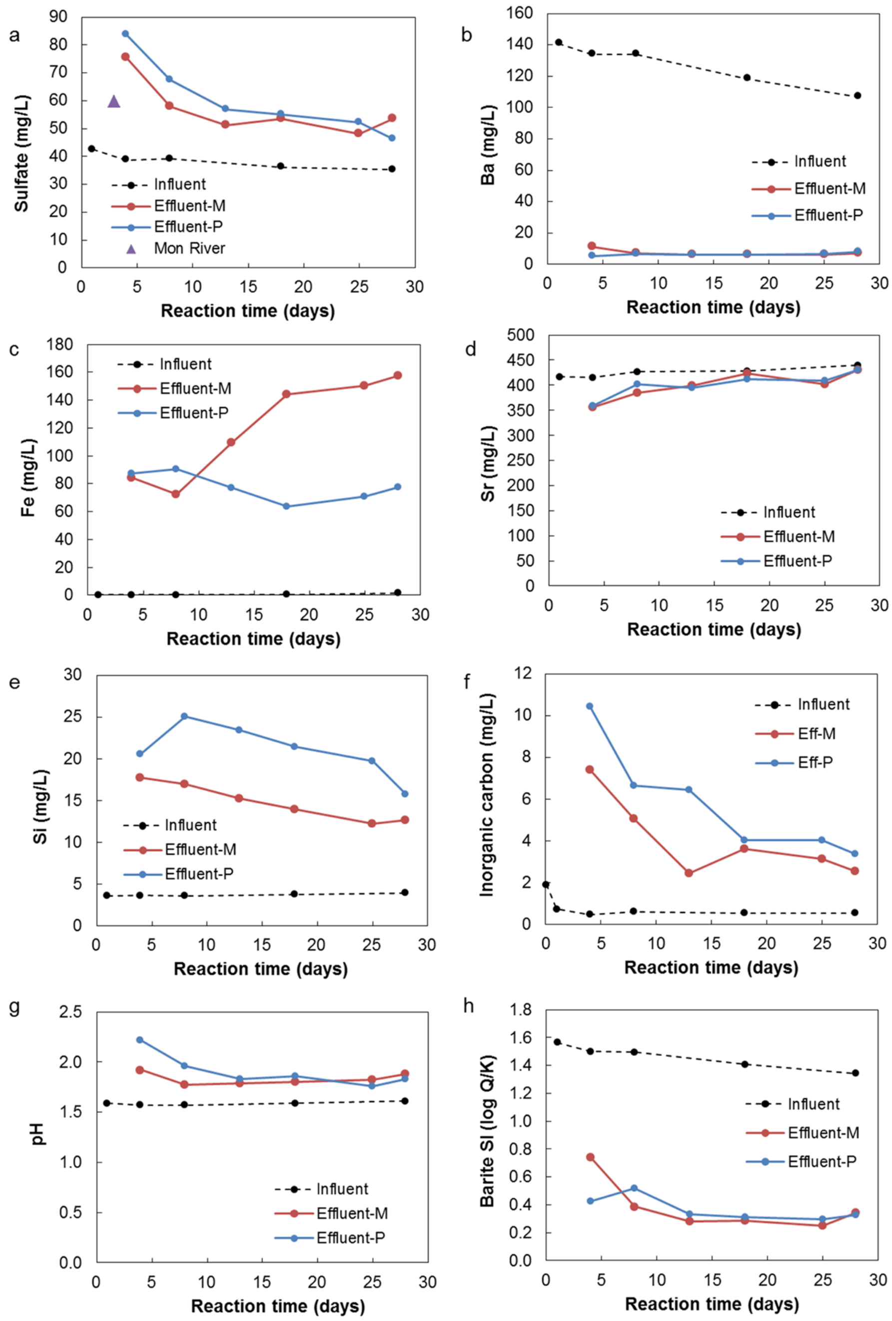


Figure 4. Selected concentrations (a−e), inorganic carbon (f), pH (g), and calculated barite saturation index (h) of the influent and effluent samples. M stands for the core with milled channels, and P stands for the core with proppants.

shows that the high Ba in the produced water may come from rock-fluid interactions by exchanging Ba^{2+} in clays with Na^+ and Ca^{2+} in the reused produced water, and barite solubility and dissolution kinetics are increased under high ionic strength conditions.[39] Another study suggests that drilling mud, which contains two orders of magnitude more barite than Ba-rich oil/gas shales, can be the major source of both barium and sulfate in hydraulic fracturing systems.[11] Sr concentration decreased a little in the effluent (Figure 4d). In this study, the Ba in barite precipitations primarily came from the injecting fluid. No

https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

detectable barite or other Ba-containing minerals were found in the Marcellus shale cores before reaction (Table 1). Very limited Ba was observed in XRF maps of the pre-reaction fracture surfaces (Figure S3).

Sulfate concentrations in the effluents were higher than that in the influent, meaning more sulfate was released than consumed by barite precipitation. Fe also increased in the effluent (Figure 4c). Pyrite oxidative dissolution contributed to the increase of sulfate and Fe. Oxidized pyrite grains were found after reaction in both our previous study[15] and the current study, while the pyrite in pre-reaction shale was unaltered without detectable oxygen in SEM−EDX (see the solid analysis below). The decomposition of persulfate was another source of sulfate. At elevated temperature (>50 °C), persulfate decomposes to a sulfate radical ($SO_4^{-}\cdot$), a strong one-electron oxidant,[40] which can rapidly degrade the guar-based polymer by defragmenting the polymer into shorter molecules, thus reducing the viscosity.[41]

The TDS in the effluent remained at similar levels present in the injection fluid (in the range of 25,000−27,000 mg/L). An aqueous Si increase (Figure 4e) indicated clay and quartz dissolution from the shale matrix. Clay dissolution is also supported by the Al increase (Figure S1). The Si concentration from the P-core was larger than that from the M-core. This could be due to larger reactive surface area, thus the Si-rich shale matrix, in the propped fracture than the milled channels exposed to the HFF. However, the Fe concentration was larger in the M-core than in the P-core, and it had an increasing trend over time. This may be because pyrite was not homogeneously distributed in the shale. As the surface minerals dissolved in the acidic HFF, it was possible that more pyrite grains were exposed on the reactive surface in the M-core than the P-core. The increase of inorganic carbon (Figure 4f) in the effluent reflects carbonate mineral dissolution when the acidic HFF flowed through the cores.

Most of the aqueous species concentrations in the effluents decreased over time. This may result from limited reactivity of fluids in the fracture with deeper fresh shale minerals below the fluid-rock interface as the reaction time increased. The pH in both effluents increased (Figure 4g) as a result of mineral dissolution to buffer the acidic HFF exposure. The injection HFF was supersaturated with respect to barite (Figure 4h), and small barite seeds were present. The barite saturation index decreased significantly in the effluent. Elevated temperature and pressure could promote barite precipitation by increasing nucleation and crystal growth rate[42] once the fluid was injected into the shale cores.

3.1.1. Barite Precipitation and Pyrite Dissolution. To quantify barite precipitation and pyrite oxidative dissolution, mass balance calculations were conducted based on the fluid chemistry data and the flow rate. Because very limited Ba would be released from the core as stated above, it is reasonable to assume that the decreased barium in the effluent turned into barite (barite Ba^{2+}). Sulfate concentrations in the effluent resulted from the completing processes. Reactions happened inside the cores, including pyrite oxidative dissolution, other minor sulfate mineral dissolution, and persulfate thermal decomposition, provided new sulfate. Since it was difficult to estimate how much persulfate was decomposed, the sulfate species generated inside the cores were grouped together as core SO_4^{2-} and barite precipitation consumed sulfate (barite SO_4^{2-}). The molar mass balance for the entire reaction duration is listed below.

$$\text{influent } Ba^{2+} + \text{barite } Ba^{2+} = \text{effluent } Ba^{2+}$$

$$\text{barite } Ba^{2+} = \text{barite } SO_4^{2-}$$

$$\text{influent } SO_4^{2-} + \text{core } SO_4^{2-} + \text{barite } SO_4^{2-} = \text{effluent } SO_4^{2-}$$

The mass balance calculation results are listed in Table 5. Influent and effluent Ba^{2+} and SO_4^{2-} were calculated based on the measured concentrations. The sulfate and barium ratio was not stoichiometric with respect to barite in the influent, with sulfate being the limiting species for barite precipitation. Sulfate and barium net changes were the difference between the effluent and influent. Sulfate net change was positive, meaning more sulfate was released inside the core than consumed. Barium net change was negative, turning into solid barite inside the core. Comparing the influent and effluent Ba mass, more than 93% barium precipitated as barite inside the cores. Compared to the influent sulfate, an appreciable amount of additional sulfate was generated inside the core (core SO_4^{2-}) from mineral dissolution and chemical decomposition. During the slow-flow shut-in period, pyrite in shale could be the primary sulfate source, other than sulfate directly from the fluid, which plays a more important role during the fast-flow stimulation period.[15] Barium became the limiting species for barite precipitation inside the cores with an ample sulfate supply during rock-fluid interactions. Limiting the barium concentration in the injection HFF can be an important step to control barite formation during the slow-flow shut-in period of hydraulic fracturing in shale.

Table 5. Mass Balance Calculation for Barium and Sulfate for 28 Days of Reaction

mass (mmol)	sulfate	sulfate net change	barium	barium net change	core SO_4^{2-}
influent	0.32		0.73		
inside core-M		0.52		−0.68	1.20
effluent-M	0.84		0.05		
inside core-P		0.61		−0.69	1.30
effluent-P	0.93		0.04		

3.2. Barite Distribution in Flow Pathways.

Barite precipitates were found in both the main flow pathways and stagnant zones (Figure 5). In the SEM/BSE images, silicate minerals in the shale matrix appeared to be darker than barite and pyrite. Barite contains heavier elements than pyrite and backscatters electrons more strongly, so barite appears to be brighter than pyrite in a grayscale. In the post-reaction cores, barite size and amount are much larger than pre-existing pyrite in shale. Barite precipitates stand out obviously (the brighter regions) in the SEM images. In the P-core, large new cracks were observed inside the core in the CT images before and after reaction. These fractures may be a result of uneven pressure on the rock surface in the presence of proppants when loading the core into the sleeve and applying confining pressure around the core holder (Figure 6a). The proppants created and expanded a new reactive surface for rock-fluid interactions in shale. No obvious new fractures formed in the M-core.

3.2.1. Barite Coating on Main Fluid Pathways. Barite coating was found in the channels of the M-core after reaction, with more barite on the bottom non-patterned side due to

https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

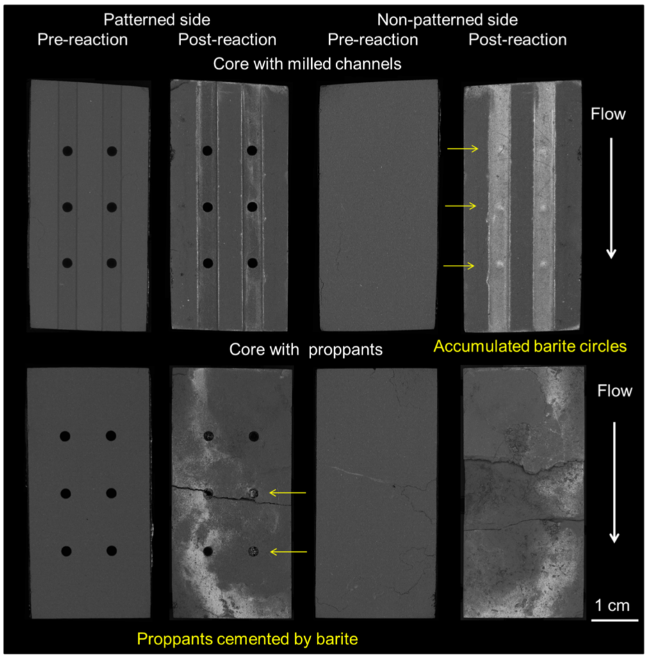


Figure 5. SEM-BSE maps of pre-reaction and post-reaction shale surfaces.

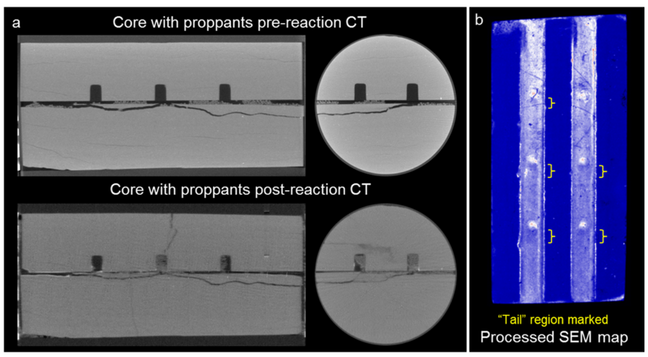


Figure 6. (a) CT images of the core with proppants before and after reaction including a side view of the left three holes and a top view of the middle two holes. Bright coating on the post-reaction fracture surface is barite. (b) Processed SEM map of the core with milled patterns, taken from the non-patterned surface, where a lighter color indicates the presence of barite. Gaussian blur and Unionjack LUT were applied on the original image using ImageJ to reduce noise and highlight gradients.

gravity. Barite seeds settled and continued to grow as the supersaturated fluid went through the channels. Figure 6b shows a processed SEM image to better show the barite distribution. The barite particle extent showed a decreasing trend in the flow direction on the surface of the channels, with more barite near the flow inlet and fewer barite near the outlet, which may be due to reduced barite seeds and Ba^{2+} supply from the fluid as the fluid-rock interactions occur along the channels. On the bottom non-patterned surface, accumulated barite spots were found at the same locations of the dead-end holes on the other side. These accumulated barite circles have a "tail" region of fewer barite behind at downstream (marked with yellow braces in Figure 6b). Some of the tail regions were more obvious than others. For laminar flow, the velocity would decrease in the hole regions. The slowed supersaturated fluid allowed more barite to settle and grow on the non-patterned bottom surface. Because barite accumulated in these circles, when the fluid came out of the hole regions, the concentration of barite seeds and Ba/sulfate supply in the fluid were lowered compared to the fluid when it came into the hole regions. Fewer barite would precipitate in the "tail" regions. When the "tail" fluid joined the main flow away from this region again, the barite seed concentration and Ba/sulfate supply increased, and the barite precipitation distribution went back to behave like in the main flow pathways with their own decreasing trend along the channel.

Barite precipitation, location, and amount are directly related to the flow access and magnitude.[15] The barite distribution on the fracture surface can in turn affect flow pathways in the propped fracture. Barite coatings were found on the surface of both sides of the P-core (Figure 5). The flow was more turbulent in the presence of proppants compared to the milled channels. The location of barite precipitation was symmetric on the top and bottom halves of the core. Barite precipitated mostly on one side of the fracture (left or right), indicating that more fluid went through that side. It is unclear as to whether this is because the core was slightly rotated inside the core holder (fracture surface was not exactly horizontal) and/or if more proppant propped the fracture wider on that side.

The initially flat surface was cracked and roughened by the proppants under confining pressure (Figure 5). The uneven pressure might trigger the pre-existing tiny fractures in the core to crack. The formation of fresh cracks was not observed in the milled channeled core without proppants.

3.2.2. Mineral Chemistry and Morphology. The pyrite grains and framboids in the shale before reaction were mostly intact with only Fe and S in the EDS spectra (Figure S4). After reaction, broken and scattered pyrite framboids were found on the fracture surface (Figure 7a,e). Much of the pyrite in the flow paths was oxidized with an appreciable amount of oxygen, as shown in the EDS spectra (Figure S4). Pyrite oxidative dissolution would provide additional sulfate to the reacted fluids. Barite precipitates were primarily $BaSO_4$, with some Sr substitutions in barite grains (Figure S5). Sr is used as a geochemical tracer of fluid-rock reactions in unconventional reservoirs.[43] Some barite contained more Sr, and some contained less. This is consistent with findings from a previous study that the distribution of Sr in barite grains was not homogeneous, even in the same barite grain.[44] This variance in Sr substitution is likely due to a shift of the crystal growth mechanism when the aqueous Sr/Ba ratio is above 5.[45] Considering that the Sr/Ba ratio went from ∼3 in the influent to ∼70 in the effluent, the localized fluid Sr/Ba ratio in the fluid pathways may be different enough to introduce heterogeneous Sr substitutions.

Barite precipitates on the shale fracture surface varied in size and morphology along the flow pathways in the experimental results. Many of the barite precipitates were small (<5 μm) and irregularly shaped. These may be the barite seeds that settled on the fracture surface from supersaturation of the injection fluid. Some barite seeds were clustered with the gelling agents in the HFF and settled together on the surface (Figure 7b). Alternatively, euhedral barite crystals (mostly 10−30 μm) were found in barite coated regions (Figure 7c). These barite most likely formed due to continued crystal growth during the course of the experimental reaction. Some barite even grew into tiny, pre-exiting fractures on the shale surface (Figure 7d). The barite seeds and euhedral barite crystals were consistent

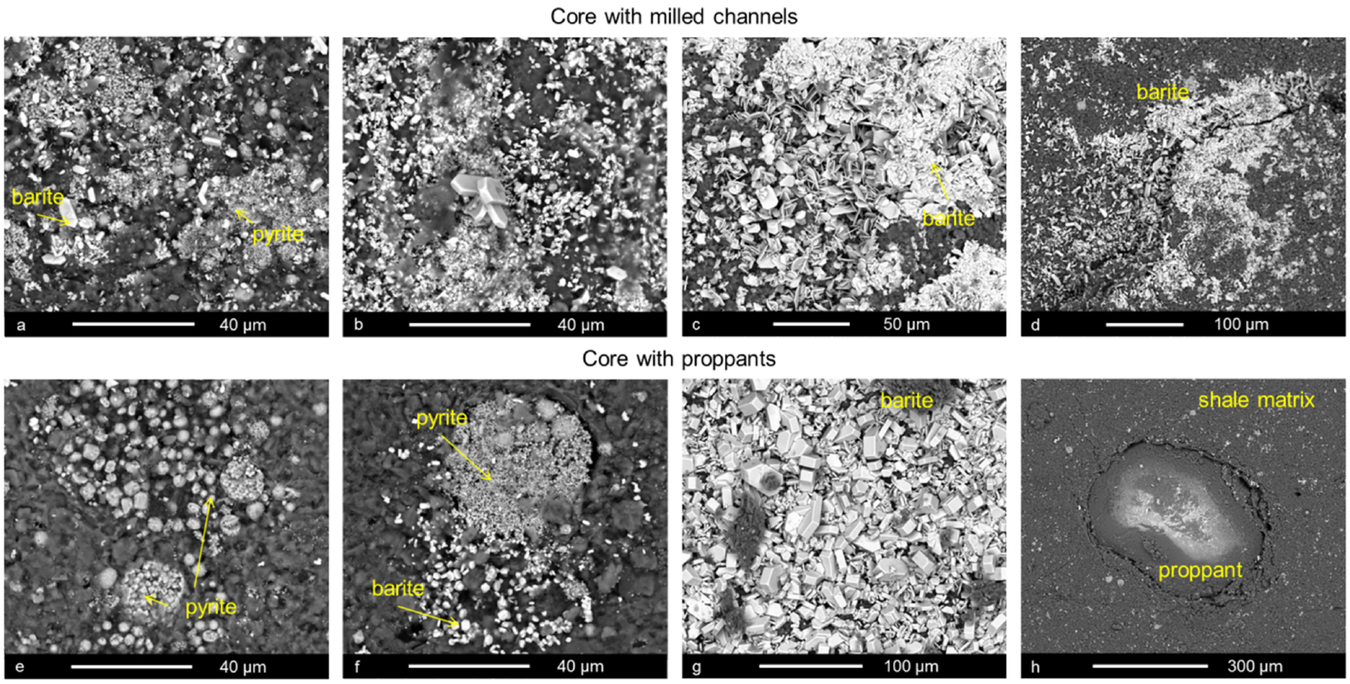


Figure 7. SEM-BSE images of minerals on the reacted core surfaces. (a−d) Core with milled channels. (e−h) Core with proppants.

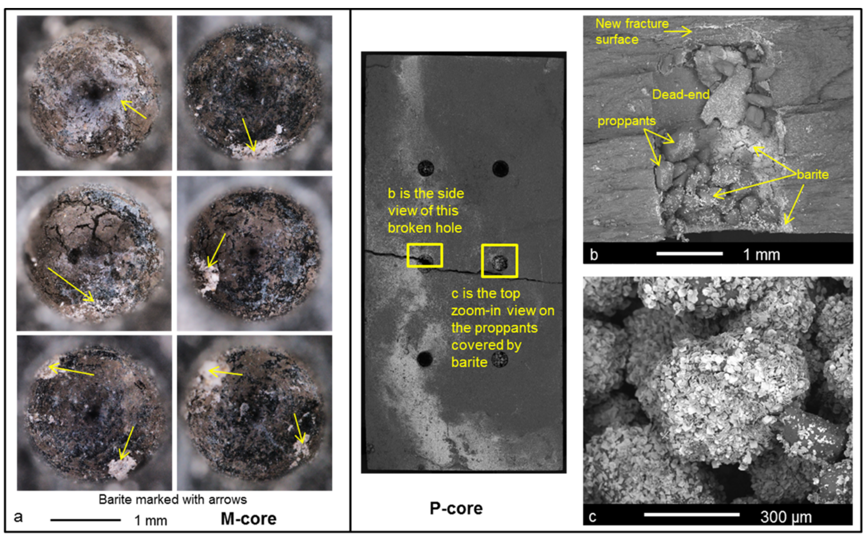


Figure 8. (a) Microscope images of the holes of the M-core, hole locations corresponding to Figure 4. White precipitates are barite; barite growth connecting proppants in dead-end holes in the P-core, (b) side view of the parted hole, (c) proppants cemented together by barite in the hole.

with the observations in our previous study.[15] Another study pointed out that euhedral barite predominantly formed at high temperature (>40 °C) and higher solution Sr/Ba ratio (>15).[46] Euhedral barite growth was expected in this experiment because the temperature was 66 °C and the Sr/Ba ratio in the solution was very high especially when considering Ba uptake by barite precipitation.

Oxidized pyrite, barite seeds, and euhedral barite were also observed in the P-core (Figure 7e,−g). No clear association between the locations of barite and pyrite was observed in both cores. In some places, barite formed around the pyrite or covered a large reactive surface containing pyrite minerals. In some places, some scattered barite was found around pyrite. In some places, no barite was around oxidized pyrite. The sulfate released from pyrite oxidative dissolution may travel with the flow and participate in barite precipitation reactions away from the origin. Sulfate from persulfate decomposition could also directly react with Ba in the fluid to form barite. Indented proppants were found on the fracture surface (Figure 7h). The proppant indentation occurred probably because radial confining pressure was applied on the core or because the shale matrix near the proppants was weakened due to dissolution in HFF exposure.[47]

3.2.3. Barite Growth in Stagnant Zones. Barite precipitation accumulated in circular patterns on the flat fracture surface opposite to the holes on the milled side of the M-core (Figure 5). These circles of barite appear to be a result of barite seeds settling directly from the HFF and continued barite growth in the slowed supersaturated flow in the hole zones. Barite precipitates were also found at the bottom of the

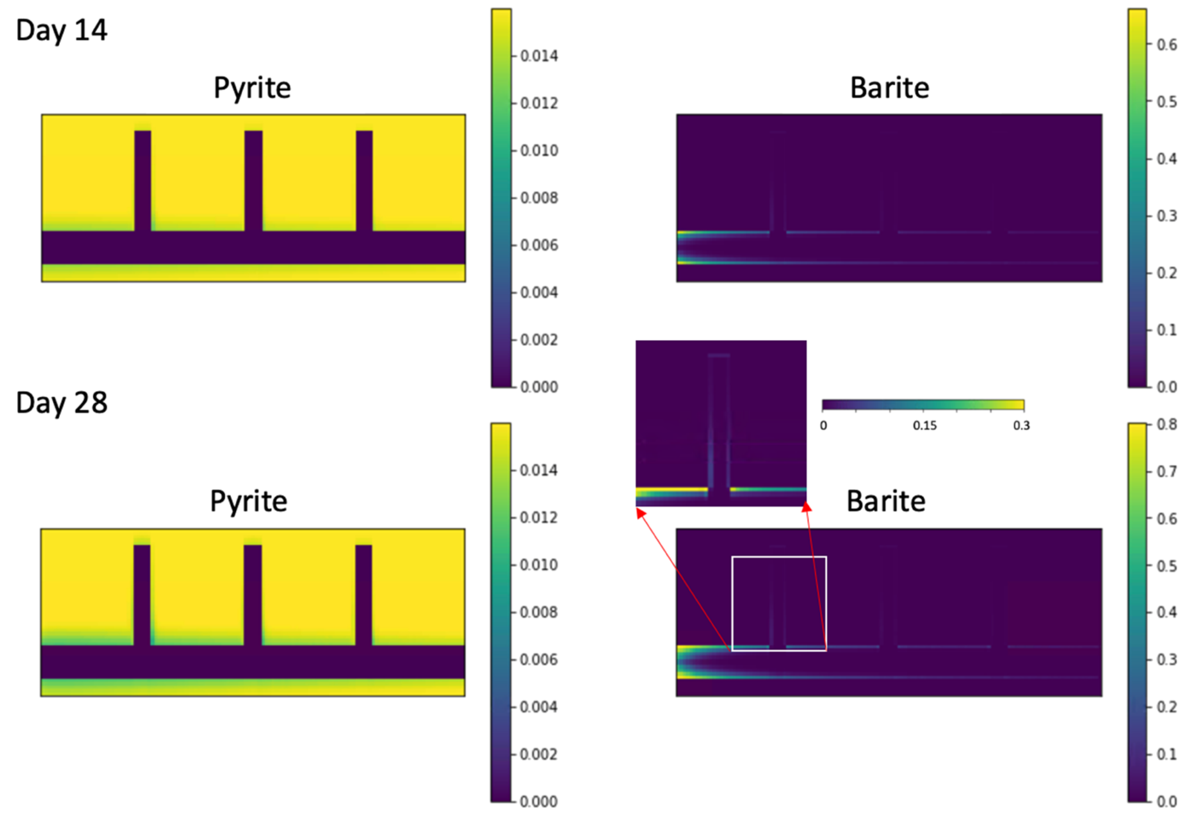


Figure 9. Volume fraction of pyrite (left) and barite (right) at day 14 (top panel) and day 28 (bottom panel).

holes on the patterned side of the M-core (Figure 8a). The patterned side was placed on top of the non-patterned side in the core holder during the course of the experiments, so these are not due to settling. Barite growth from the stagnant fluid led to the scaling on the inner surface of the holes. Continuous Ba and sulfate supply from the hole opening from the laminar flow also contributed to barite growth in the stagnant zones.

In the P-core, the proppants inside the dead-end holes were cemented together into larger accretions by barite (Figure 8b,c). The cemented proppants remained stable inside these holes even with the non-patterned side facing down. The patterned side was on top of the non-patterned side, so fewer small barite seeds were settled there due to gravity. These euhedral barite crystals formed due to geochemical reactions within the core. In a flow-through study using glass beads as porous media, barite precipitations were found to cover the bead surfaces and fill void space in between the beads.[48] In contrast, proppants located in the main flow pathways were rarely found to be covered with barite (Figure 7h). Barite also grew further into new crack surfaces (Figure 8b).

The stagnant zones can be viewed as porous media after the proppants were flushed into the dead-end holes. The minimal flow velocity in the stagnant zones allowed barite to precipitate with less turbulent supersaturated fluid and the continuous supply of Ba and sulfate from the main flow at the opening of the hole in diffusion-controlled transport zones. Barite scaling and proppant cementation can be an issue for shale gas production if they block the transport pathways in shale and/or pipelines.

3.3. Reactive Transport Modeling. Reactive transport modeling showed a decreased pyrite volume fraction within the shale matrix over time (Figure 9). Sulfate was predicted to be present in the fluid due to pyrite oxidative dissolution (Figure S6), which was consistent with the presence of sulfate in the effluent from the experiment. The amount of pyrite dissolution was not as significant as observed in the experiment because only dissolved oxygen under atmospheric conditions was included in the simulation, while the actual fluid contained other oxidants such as persulfate. The amount of barite precipitation in the fracture increased over time, and the extent of barite coating on the channel surface decreased along the flow direction (Figure 9). The decreasing barite gradient along the flow direction was because of decreasing supersaturation following the consumption of Ba^{2+} from the fluid. A similar gradient along the channels was observed in the experiment (Figure 6b).

The total barite precipitation within each column of grid cells is shown in Figure 10, assuming a thickness of 1 mm for the 2D cross section. The barite precipitation amount close to the inlet was one order of magnitude higher than that at the outlet. Barite precipitation hotspots (the spikes) were also

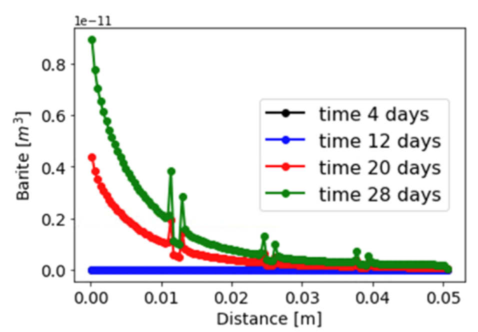


Figure 10. Amount of barite precipitation along the flow direction. The spikes are in the dead-end regions.

https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

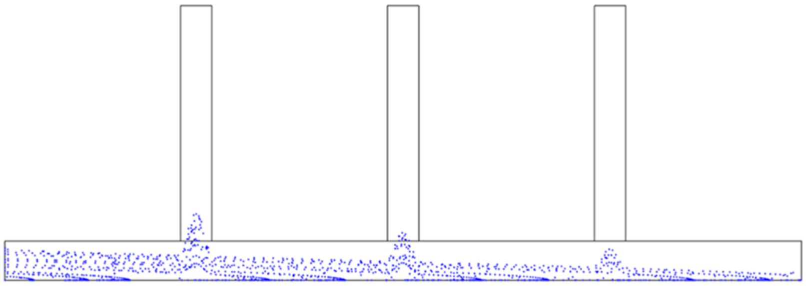


Figure 11. Particle (blue dots) transport in the simulated domain. The y axis is exaggerated by 5 to better show the particles.

observed around the dead-ends. The walls of the dead-ends provided additional surface area for the precipitation reaction in the simulation.

Based on the pressure difference (ΔP) recorded in the reactive transport modeling, fracture permeability (k) can be evaluated using $k = \frac{\overline{u}\mu L}{\Delta P}$. Due to barite precipitation, the fracture permeability decreased by ~16% from the initial value of 22 Darcy to ~18 Darcy. Another common practice is to estimate fracture permeability based on average aperture ($\overline{b}$) using the cubic law, $k = \frac{\overline{b}^2}{12}$. The amount of barite precipitation corresponded to ~5% reduction in the average aperture and thus ~10% in the permeability. This difference is because precipitation and thus aperture change were not uniform along the flow direction. More precipitation was observed at the inlet and created a stronger flow restriction. It further highlights the importance of the spatial distribution of the precipitates in controlling fracture permeability change.

3.3.1. The Transport of Colloidal Barite. The reactive transport modeling did not consider particle transport and deposition and thus the impacts of colloidal barite. Colloidal barite formed in the injection fluid before entering into the fracture domain. Once the fluid was injected, the colloidal barite would act as the seed for further crystal growth. Because of gravity, the colloidal barite settled on the bottom fracture surface, providing the additional surface area for continuous barite growth. In the reactive transport simulation, all fracture surface area was considered equal for precipitation, whereas in reality, precipitation on existing barite surfaces was more energetically favorable. In the experiment, barite precipitation was primarily observed on the fracture surface of the bottom half core, on which barite settled primarily.

Figure 11 illustrates particle settling distribution within the fracture domain for the flow field given in Figure 3. The simulation was performed in the COMSOL Multiphysics code using the "particle tracing for fluid flow" application. The particle size was set to 5 μm, and the density was 1050 kg/m^3, and 10 particles were released every second at the inlet. The particles settled on the bottom of the channels with a decreasing extent along the flow direction. Accumulated barite particles were found at the openings of the dead-end regions, which was consistent with the experimental observations (Figure 6b).

3.3.2. Impact of Reactive Flow on Scaling in Different Pathways. Barite distribution in channeled fracture vs propped fracture and stagnant zones vs main flow regions was investigated. In the channeled fracture, the flow was close to laminar flow. Barite settled and grew along the channel surface with a decreasing extent trend. Such a barite distribution trend was not obvious in the propped fracture, in which the flow was turbulent. Barite distribution was associated with the flow access and magnitude in the propped fracture.[15] The distribution of barite in the main flow regions and stagnant zones was a result of the coupling of reactive transport and particle settling. The barite formed in the injection fluid settled in the main flow pathways and served as seeds for continuous barite growth in the reactive flow. In the particle transport simulation without geochemical reactions, no barite was predicted on the wall of the dead-ends, which were placed on the top (Figure 11). In reactive transport modeling, stagnant fluids in the dead-ends were supersaturated and barite could form on the surface in these regions (Figure 9), as observed in the experiment (Figure 8). This indicates that even without settled barite seeds to facilitate crystal growth, barite can still form in regions where the supersaturated fluid is stagnant. Our previous study with Marcellus shale powders showed continued barite precipitation due to rock-fluid interactions after removing the pre-precipitated barite in the initial HFF, even with a scale inhibitor.[24] Apart from the main flow pathways for barite to precipitate, these dead-end regions also allow barite precipitation once the supersaturated fluid was injected into the reservoir. More importantly, barite could connect and cement the proppants that were trapped in such stagnant zones and form larger conglomerates. Such barite-proppant conglomerates were not found in the main flow pathways in the P-core. Shale mineral dissolution may increase porosity and permeability for gas extraction. However, the barite coatings on the reactive surfaces and barite-proppant conglomerates could inhibit mass transport and mineral dissolution in the rock-fluid interface.

4. CONCLUSIONS

Colloidal barite in the injection HFF can settle in the main flow pathways and serve as a seed for continuous barite growth during the entire fracturing process. Barite coatings can cover a large portion of the reactive surface in shale, blocking transport pathways in the rock-fluid interface. Removing pre-precipitated barite before injection may help control barite scaling once the fluid is injected into the reservoir. Barite may still continue to form due to rock-fluid interactions in the reservoir. Supersaturated fluids allow barite to precipitate in stagnant zones. With the trapped proppants in the stagnant zones, individual proppants can be cemented together by barite precipitations and become bulk barite-proppant conglomerates, inhibiting or even block the transport pathways. Proppants conglomerating via barite scaling are a potential issue associated with scale control in the shale gas industry.

https://doi.org/10.1021/acs.energyfuels.1c01374
Energy Fuels 2021, 35, 11947−11957

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.energyfuels.1c01374.

> Other major cation and anion concentrations; XRF elemental maps of the fracture surfaces of the pre-reaction cores; SEM–EDX of pyrite in pre- and post-reaction shale; SEM image of barite with Sr substitution; model predicted effluent concentrations (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Wei Xiong – *National Energy Technology Laboratory, Pittsburgh, Pennsylvania 15236, United States; NETL Support Contractor, Pittsburgh, Pennsylvania 15236, United States;* orcid.org/0000-0002-0793-3810; Phone: 412-386-7324; Email: wei.xiong@netl.doe.gov

Authors

Hang Deng – *Lawrence Berkeley National Laboratory, Berkeley, California 94720, United States;* orcid.org/0000-0001-5784-996X

Johnathan Moore – *National Energy Technology Laboratory, Morgantown, West Virginia 26505, United States; NETL Support Contractor, Morgantown, West Virginia 26505, United States*

Dustin Crandall – *National Energy Technology Laboratory, Morgantown, West Virginia 26505, United States*

J. Alexandra Hakala – *National Energy Technology Laboratory, Pittsburgh, Pennsylvania 15236, United States;* orcid.org/0000-0003-3325-6160

Christina Lopano – *National Energy Technology Laboratory, Pittsburgh, Pennsylvania 15236, United States*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.energyfuels.1c01374

Notes

The authors declare no competing financial interest.

This project was funded by the United States Department of Energy, National Energy Technology Laboratory, in part, through a site support contract. Neither the United States Government nor any agency thereof, nor any of their employees, nor the support contractor, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of any information, apparatus, product, or process disclosed, or represents that its use would not infringe privately owned rights. Reference herein to any specific commercial product, process, or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply its endorsement, recommendation, or favoring by the United States Government or any agency thereof. The views and opinions of authors expressed herein do not necessarily state or reflect those of the United States Government or any agency thereof.

■ ACKNOWLEDGMENTS

This study was supported by the U.S. Department of Energy's (DOE) Office of Fossil Energy as part of the National Energy Technology Laboratory's (NETL) ongoing research, and partly by appointments to the NETL research Participation Program, sponsored by the U.S. DOE and managed by the Oak Ridge Institute for Science and Education (ORISE). Thanks to the people who assisted us: the X-ray CT scan and technical team at NETL Morgantown and the analysts at NETL Pittsburgh Analytical Laboratory. Thanks to Josh Miller on IC analysis. We appreciate valuable advice and discussions with Mengling Stuckman, Qingyun Li, Adam Jew, and John Bargar. Any use of trade, product, or firm names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

■ REFERENCES

(1) Chen, H.; Carter, K. E. Characterization of the chemicals used in hydraulic fracturing fluids for wells located in the Marcellus Shale Play. *J. Environ. Manage.* **2017**, *200*, 312–324.

(2) U.S. Environmental Protection Agency *Analysis of Hydraulic Fracturing Fluid Data from the FracFocus Chemical Disclosure Registry 1.0. Office of Research and Development*; U.S. Environmental Protection Agency: Washington, DC. EPA/601/R-14/003, 2015.

(3) Vidic, R. D.; Brantley, S. L.; Vandenbossche, J. M.; Yoxtheimer, D.; Abad, J. D. Impact of shale gas development on regional water quality. *Science* **2013**, *340*, 1235009.

(4) Kerr, R. A. Natural Gas From Shale Bursts Onto the Scene. *Science* **2010**, *328*, 1624–1626.

(5) McGlade, C.; Speirs, J.; Sorrell, S. Methods of estimating shale gas resources – Comparison, evaluation and implications. *Energy* **2013**, *59*, 116–125.

(6) Karra, S.; Makedonska, N.; Viswanathan, H. S.; Painter, S. L.; Hyman, J. D. Effect of advective flow in fractures and matrix diffusion on natural gas production. *Water Resour. Res.* **2015**, *51*, 8646–8657.

(7) Patzek, T. W.; Male, F.; Marder, M. Gas production in the Barnett Shale obeys a simple scaling theory. *Proc. Natl. Acad. Sci. U. S. A.* **2013**, *110*, 19731–19736.

(8) Jew, A. D.; Dustin, M. K.; Harrison, A. L.; Joe-Wong, C. M.; Thomas, D. L.; Maher, K.; Brown, G. E.; Bargar, J. R. Impact of Organics and Carbonates on the Oxidation and Precipitation of Iron during Hydraulic Fracturing of Shale. *Energy Fuels* **2017**, *31*, 3643–3658.

(9) Kargbo, D. M.; Wilhelm, R. G.; Campbell, D. J. Natural Gas Plays in the Marcellus Shale: Challenges and Potential Opportunities. *Environ. Sci. Technol.* **2010**, *44*, 5679–5684.

(10) Akob, D. M.; Cozzarelli, I. M.; Dunlap, D. S.; Rowan, E. L.; Lorah, M. M. Organic and inorganic composition and microbiology of produced waters from Pennsylvania shale gas wells. *Appl. Geochem.* **2015**, *60*, 116–125.

(11) Jew, A. D.; Li, Q.; Cercone, D.; Maher, K.; Gordon, E.; Brown, J.; Bargar, J. R. Barium Sources in Hydraulic Fracturing Systems and Chemical Controls on its Release into Solution. In *Unconventional Resources Technology Conference (URTeC)*; Society of Petroleum Engineers: Houston, Texas, USA, 2018.

(12) Abdelgawad, K.; Mahmoud, M.; Elkatatny, S.; Patil, S. Effect of Calcium Carbonate on Barite Solubility Using a Chelating Agent and Converter. In *SPE International Conference on Oilfield Chemistry*; Society of Petroleum Engineers: Galveston, Texas, USA, 2019.

(13) He, C.; Vidic, R. D. Impact of Antiscalants on the Fate of Barite in the Unconventional Gas Wells. *Environ. Eng. Sci.* **2016**, *33*, 745–752.

(14) Hakala, J. A.; Crandall, D.; Moore, J.; Phan, T.; Sharma, S.; Lopano, C. Laboratory-Scale Studies on Chemical Reactions Between Fracturing Fluid and Shale Core from the Marcellus Shale Energy and Environmental Laboratory (MSEEL) Site. In *Unconventional Resources Technology Conference (URTEC)*; Society of Petroleum Engineers: Austin, Texas, USA, 2017.

(15) Xiong, W.; Gill, M.; Moore, J.; Crandall, D.; Hakala, J. A.; Lopano, C. Influence of Reactive Flow Conditions on Barite Scaling in Marcellus Shale during Stimulation and Shut-In Periods of Hydraulic Fracturing. *Energy Fuels* **2020**, *34*, 13625–13635.

(16) Wilson, J. M.; Van Briesen, J. M. Source water changes and energy extraction activities in the Monongahela River, 2009-2012. *Environ. Sci. Technol.* **2013**, *47*, 12575−12582.

(17) World Health Organization *Sulfate in Drinking-water*; World Health Organization: 2004.

(18) Chapman, E. C.; Capo, R. C.; Stewart, B. W.; Kirby, C. S.; Hammack, R. W.; Schroeder, K. T.; Edenborn, H. M. Geochemical and strontium isotope characterization of produced waters from Marcellus Shale natural gas extraction. *Environ. Sci. Technol.* **2012**, *46*, 3545−3553.

(19) Flynn, S. L.; von Gunten, K.; Warchola, T.; Snihur, K.; Forbes, T. Z.; Goss, G. G.; Gingras, M. K.; Konhauser, K. O.; Alessi, D. S. Characterization and implications of solids associated with hydraulic fracturing flowback and produced water from the Duvernay Formation, Alberta, Canada. *Environ. Sci.: Processes Impacts* **2019**, *21*, 242−255.

(20) Balaba, R. S.; Smart, R. B. Total arsenic and selenium analysis in Marcellus shale, high-salinity water, and hydrofracture flowback wastewater. *Chemosphere* **2012**, *89*, 1437−1442.

(21) Haluszczak, L. O.; Rose, A. W.; Kump, L. R. Geochemical evaluation of flowback brine from Marcellus gas wells in Pennsylvania, USA. *Appl. Geochem.* **2013**, *28*, 55−61.

(22) He, C.; Li, M.; Liu, W.; Barbot, E.; Vidic, R. D. Kinetics and Equilibrium of Barium and Strontium Sulfate Formation in Marcellus Shale Flowback Water. *J. Environ. Eng.* **2014**, *140*, B4014001.

(23) Rassenfoss, S. From flowback to fracturing: water recycling grows in the Marcellus shale. *J. Pet. Technol.* **2011**, *63*, 48−51.

(24) Xiong, W.; Lopano, C.; Hakala, A.; Carney, B. Investigation of Barite Scaling During Reaction between Pre-Treated Hydraulic Fracturing Fluid from the Field and Marcellus Shale. In *Unconventional Resources Technology Conference (URTEC)*; OnePetro: Austin, Texas, USA, 2020.

(25) Li, Q.; Jew, A. D.; Kohli, A.; Maher, K.; Brown, G. E.; Bargar, J. R. Thicknesses of Chemically Altered Zones in Shale Matrices Resulting from Interactions with Hydraulic Fracturing Fluid. *Energy Fuels* **2019**, *33*, 6878−6889.

(26) Paukert Vankeuren, A. N.; Hakala, J. A.; Jarvis, K.; Moore, J. E. Mineral Reactions in Shale Gas Reservoirs: Barite Scale Formation from Reusing Produced Water As Hydraulic Fracturing Fluid. *Environ. Sci. Technol.* **2017**, *51*, 9391−9402.

(27) Moore, J.; Hakala, A.; Lopano, C. L.; Xiong, W.; Phan, T.; Vankeuren, A.; Sharma, S.; Pilewski, J.; Jarvis, K.; Brown, S. In *Bench-Top Experiments Evaluating Simulated Hydraulic Fracturing Fluid Interactions With Marcellus Shale Core, Unconventional Resources Technology Conference, 2018*; Society of Exploration Geophysicists: 2018.

(28) Li, Q.; Jew, A. D.; Kiss, A. M.; Kohli, A.; Alalli, A.; Kovscek, A. R.; Zoback, M. D.; Cercone, D.; Maher, K.; Gordon, E.; Brown, J.; Bargar, J. R. Imaging Pyrite Oxidation and Barite Precipitation in Gas and Oil Shales. In *Unconventional Resources Technology Conference (URTeC)*; Society of Exploration Geophysicists: Houston, Texas, USA, 2018.

(29) Harrison, A. L.; Jew, A. D.; Dustin, M. K.; Thomas, D. L.; Joe-Wong, C. M.; Bargar, J. R.; Johnson, N.; Brown, G. E.; Maher, K. Element release and reaction-induced porosity alteration during shale-hydraulic fracturing fluid interactions. *Appl. Geochem.* **2017**, *82*, 47−62.

(30) Godinho, J. R. A.; Ma, L.; Chai, Y.; Storm, M.; Burnett, T. L. Mineral Precipitation in Fractures and Nanopores within Shale Imaged Using Time-Lapse X-ray Tomography. *Minerals* **2019**, *9*, 480.

(31) Phan, T. T.; Hakala, J. A.; Lopano, C. L.; Sharma, S. Rare earth elements and radiogenic strontium isotopes in carbonate minerals reveal diagenetic influence in shales and limestones in the Appalachian Basin. *Chem. Geol.* **2019**, *509*, 194−212.

(32) Mouser, P. J.; Liu, S.; Cluff, M. A.; McHugh, M.; Lenhart, J. J.; MacRae, J. D. Redox conditions alter biodegradation rates and microbial community dynamics of hydraulic fracturing fluid organic additives in soil−groundwater microcosms. *Environ. Eng. Sci.* **2016**, *33*, 827−838.

(33) Hammack, R.; Harbert, W.; Sharma, S.; Stewart, B.; Capo, R.; Wall, A. *An evaluation of fracture growth and gas/fluid migration as horizontal Marcellus Shale gas wells are hydraulically fractured in Greene County, Pennsylvania*; National Energy Technology Laboratory: NETL-TRS-3-2014, 76pp. 2014.

(34) Schneider, C. A.; Rasband, W. S.; Eliceiri, K. W. NIH Image to ImageJ: 25 years of image analysis. *Nat. Methods* **2012**, *9*, 671−675.

(35) Steefel, C.; Appelo, C.; Arora, B.; Jacques, D.; Kalbacher, T.; Kolditz, O.; Lagneau, V.; Lichtner, P.; Mayer, K. U.; Meeussen, J.; Molins, S.; Moulton, D.; Shao, H.; Šimůnek, J.; Spycher, N.; Yabusaki, S. B.; Yeh, G. T. Reactive transport codes for subsurface environmental simulation. *Comput. Geosci.* **2015**, *19*, 445−478.

(36) Li, Q.; Jew, A. D.; Brown, G. E., Jr.; Bargar, J. R.; Maher, K. Reactive Transport Modeling of Shale−Fluid Interactions after Imbibition of Fracturing Fluids. *Energy Fuels* **2020**, *34*, 5511−5523.

(37) Moon, T. C.; Lucostic, C. M. Effects of acid mine drainage on a southwestern Pennsylvania stream. *Water, Air, Soil Pollut.* **1979**, *11*, 377−390.

(38) U.S. Geological Survey *Geological Survey Water-supply Paper*; U.S. Government Printing Office: 1962.

(39) Renock, D.; Landis, J. D.; Sharma, M. Reductive weathering of black shale and release of barium during hydraulic fracturing. *Appl. Geochem.* **2016**, *65*, 73−86.

(40) Johnson, R. L.; Tratnyek, P. G.; Johnson, R. O. B. Persulfate persistence under thermal activation conditions. *Environ. Sci. Technol.* **2008**, *42*, 9350−9356.

(41) Al-Muntasheri, G. A. *A Critical Review of Hydraulic Fracturing Fluids over the Last Decade, SPE Western North American and Rocky Mountain Joint Meeting, 2014*; SPE Production & Operations: 2014.

(42) Tranter, M.; De Lucia, M.; Kühn, M. Numerical investigation of barite scaling kinetics in fractures. *Geothermics* **2021**, *91*, 102027.

(43) Bayless, E. R.; Bullen, T. D.; Fitzpatrick, J. A. Use of 87Sr/86Sr and δ11B to identify slag-affected sediment in southern Lake Michigan. *Environ. Sci. Technol.* **2004**, *38*, 1330−1337.

(44) Hunter, H. A.; Ling, F. T.; Peters, C. A. Metals Coprecipitation with Barite: Nano-XRF Observation of Enhanced Strontium Incorporation. *Environ. Eng. Sci.* **2020**, *37*, 235−245.

(45) Weber, J.; Bracco, J. N.; Poplawsky, J. D.; Ievlev, A. V.; More, K. L.; Lorenz, M.; Bertagni, A. L.; Jindra, S. A.; Starchenko, V.; Higgins, S. R.; Stack, A. G. Unraveling the Effects of Strontium Incorporation on Barite Growth—In Situ and Ex Situ Observations Using Multiscale Chemical Imaging. *Cryst. Growth Des.* **2018**, *18*, 5521−5533.

(46) Widanagamage, I. H.; Waldron, A. R.; Glamoclija, M. Controls on Barite Crystal Morphology during Abiotic Precipitation. *Minerals* **2018**, *8*, 480.

(47) Brown, S.; Moore, J. E.; Crandall, D. M.; Hakala, J. A.; Lopano, C. L.; Xiong, W. *Proppant embedment in shale during exposure to hydraulic fracturing fluids*; National Energy Technology Laboratoty: 2017; Vol. No. NETL-PUB-21599.

(48) Godinho, J. R.; Gerke, K. M.; Stack, A. G.; Lee, P. D. The dynamic nature of crystal growth in pores. *Sci. Rep.* **2016**, *6*, 33086.